Arizona State University
Department of Economics
Ph.D. Qualifying Exam

Macroeconomic Theory

June 3, 2019

Instructions: Read the questions carefully, show all your work, and explain all your answers. Indicate carefully if you introduce notation, or make assumptions when answering a question. Please make sure to allocate time to ALL three problems.

Good luck!

1. Consider the following asset-pricing model. The economy consists of two countries, 1 and 2. Each country is populated with a continuum of individuals of mass one. In each country $i \in \{1, 2\}$, there is a tree producing a sequence of dividends z_i. The process (z_1, z_2) is Markov with the transition function Q. Individuals in country i can only trade stocks of tree i that grows in their country, but not the other tree. In addition, all individuals in the economy can trade a one-period bond with each other.

Each individual maximizes the expected discounted utility of consumption. All individuals have the same instantaneous utility $U(c)$, where c denotes consumption, and the function U is strictly increasing, strictly concave, and continuously differentiable. Individuals discount the future with $\beta \in (0, 1)$. The number of shares of each tree is normalized to one. At $t = 0$, each person in country i is endowed with one share of tree i and $b_{i,0}$ bonds. The consumption good is not storable.

(a) Set up the recursive problems of individuals in the two countries.

(b) Define a stationary recursive competitive equilibrium in this economy.

(c) Derive first-order conditions, Envelope conditions, and Euler equations for individuals of each country.

(d) Suppose that the dividend of the first tree equals 0 in all odd periods and 1 in all even periods. The dividend of the second tree equals 1 in all odd periods and 0 in all even periods. Consider a stationary equilibrium where each individual's consumption is constant over time. Find the corresponding equilibrium prices of the bond and stocks in closed form.

(e) Economist X claims that the stock prices in the economy in part (d) are higher than they would have been in the economy with linear preferences and no bond. Is he right? Explain.

2. Consider the following version of the overlapping generations model with capital. The novelty in the current case is that there are two types of individuals in each cohort who differ only in their labor endowments at each age.

Time is discrete ($t = 0, 1, 2, ...$). A cohort of size N is born every period. There is no population growth nor growth in labor efficiency. Individuals live for two periods, and work in both periods of their lives. There are two types of individuals in each cohort. Individuals of type C (with a 'college degree') have a labor endowment of 1 in the first period of their lives, and $(1 + g)$ in their second period, with $g > 0$. Individuals of type NC (without a 'college degree') have a labor endowment of 1 in each period. Individuals of type C constitute a fraction ϕ of each cohort, while individuals of type NC constitute a corresponding fraction $1 - \phi$. This is the only source of heterogeneity among individuals.

An individual born at date t chooses consumption levels $c_{1,t}$, $c_{2,t+1}$ and savings s_t, to maximize

$$U(c_{1,t}, c_{2,t+1}) = \ln(c_{1,t}) + \beta \ln(c_{2,t+1}).$$

Competitive firms have access to a Cobb-Douglas production technology that uses capital and labor services. Capital fully depreciates within a period.

(a) Set up an individual's problem. Derive the decision rules for savings.

(b) Could there be a competitive equilibrium in which there is borrowing by some individuals? If so, who would borrow? *Intuitively*, under what conditions would such an equilibrium exist?

(c) Derive the equilibrium law of motion for the capital-labor ratio. What is the effect of a higher fraction of individuals of type C on the capital-labor ratio in steady state? Explain.

(d) A White House Economic Advisor states that because the labor endowment of type-C agents grows over their lives, even when there is no growth in population or labor efficiency, the steady state equilibrium of this economy is not dynamically efficient.

Is he right? Explain carefully.

3. Consider a discrete-time infinite-horizon model in which an agent chooses between working and not working for the rest of her life. The employment choice can only be made in period 0 and is irreversible. If the agent chooses to work, she receives income $y > 0$ in every period and has access to a risk-free saving technology paying the rate of return r. If the agent does not work, she receives no income but still has access to the same saving technology. In either case, assets cannot be negative: $a_t \geq 0$ for all $t \geq 1$. The initial asset level is $a_0 > 0$. Working is associated with disutility $\phi > 0$ in every period, so period-t utility of a working agent is $u(c_t) - \phi$, while the period-t utility of a non-working agent is $u(c_t)$, where c_t is consumption in period t and $u(\cdot)$ is a strictly increasing, strictly concave, and continuously-differentiable function defined on $\mathbb{R}_+$, such that $\lim_{c \to 0} u'(c) = +\infty$. The agent discounts future with a time discount factor $\beta \in (0, 1)$, implying that the life-time utility of the agent is

$$\sum_{t=0}^{\infty} \beta^t \left[u(c_t) - \mathbb{I}_1 \phi \right],$$

where $\mathbb{I}_1$ is an indicator function describing whether the agent chooses to work.

(a) Set up the dynamic decision problem of the agent.

(b) Under which condition(s) does the constraint $a_t \geq 0$ never bind regardless of the employment choice in period 0?

(c) Assume that the condition(s) derived in (b) above holds. Denote by $c_1(a)$ and $c_0(a)$ the policy function describing the agent's optimal consumption as a function of her current-period assets conditional on working and not working, respectively. Prove that $c_1(a) > c_0(a)$ for any a.

(d) Develop a formal argument to characterize the agent's employment choice in period 0 as a function of her initial asset level. Provide an intuitive explanation.